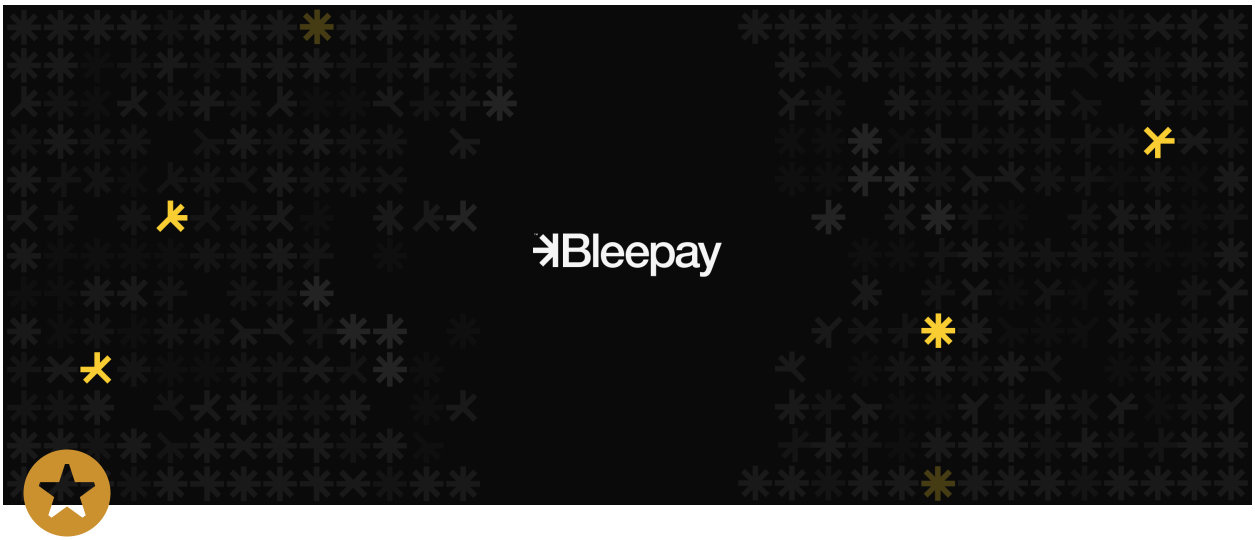


Patent

Systems and Methods for Voucher-Based Authorization and Redemption of Digital Asset Payments

Technical Field

The present invention relates to secure and efficient online payment capabilities within electronic payment networks utilizing cryptocurrencies or other forms of digital assets. More particularly, the invention concerns computer-implemented methods for transferring digital assets through a proxied voucher mechanism, wherein the processes of transfer, redemption and authorization are mediated by a short-lived voucher. This enables users to perform verified and authenticated asset transfers without directly exposing public or private keys or underlying transaction credentials, thereby enhancing security, usability and interoperability across digital asset platforms.

Background

Multiple sources describe characteristics that a cryptocurrency or digital asset platform should possess in order to promote wide adoption. One of the most

important among these characteristics is the ability to provide a smooth and intuitive user experience that drives financial inclusion on a large scale—that is, a system that can be easily used by anyone, anywhere, regardless of their familiarity with blockchain technology or the complexity of the digital tools they possess. In practice, achieving full inclusivity remains challenging. Complex key management, difficult wallet interfaces, confusing transaction flows and security-related fears continue to prevent average users from participating in the digital asset economy. Therefore, a truly inclusive cryptocurrency platform must incorporate user-friendly methods that allow secure and efficient transactions while abstracting the underlying cryptographic complexity.

While many improvements have been made in areas such as wallet usability and transaction speed, one aspect remains particularly problematic—the simplicity and reliability of authorization and asset transfer processes. In most existing systems, users are required to manage private keys or multi-step authentication procedures directly, which can be intimidating and error-prone. Furthermore, existing wallet architectures typically rely on maintaining a direct, stateful connection between the user's wallet and a decentralized application (dApp) to authorize or complete a transaction. This stateful connection model introduces friction, limits interoperability and exposes users to potential connection and synchronisation issues.

There is, therefore, a need for a mechanism by which users can transfer, redeem and authorize digital asset transactions through a simple, proxied digital voucher. Such a mechanism would allow users to conduct secure transfers within an online cryptocurrency network without directly exposing public or private keys or establishing a stateful session between a wallet and an application, while ensuring that all transactions remain verifiable, authorised and redeemable within the network. From a technical perspective, such a system must support indirect authorization and redemption while preserving cryptographic integrity. It must ensure that a digital voucher—although simple for a human to obtain, input or share—serves as a secure proxy for transaction initiation and validation. Existing online cryptocurrency payment protocols are optimized for direct key use and stateful wallet interactions, not for proxied, stateless communication. As a result, there exists a clear and growing need for a secure, voucher-based proxy protocol

capable of performing transfer, redemption and authorization operations within cryptocurrency networks.

Several existing patents describe systems that use temporary numeric codes, electronic vouchers or one-time passwords to authorize or facilitate payment transactions. These systems have proven effective in traditional financial environments but are not adapted for use within decentralized digital asset architectures.

WO 2021/058403 A1 discloses a method and system for executing a transaction that may incorporate payment schemes such as BLIK, PayPal or PSD2-compliant frameworks. While it recognizes the use of numeric codes as part of centralized payment processors, it remains bound to issuer–acquirer infrastructure. In contrast, the present invention defines a voucher that functions as an off-chain cryptographic authorization proxy, resolvable directly into a blockchain-compatible transaction object. Each voucher corresponds to a pre-structured, unsigned transaction template that can be cryptographically completed and broadcast by the payer’s wallet. The system thereby eliminates dependency on issuer–acquirer networks and transforms numeric-code authorization into a protocol-level mechanism for decentralized settlement.

EP 2299397 A1 discloses an electronic voucher system in which a six-digit code is sent by SMS to a user and redeemed at a chip-and-PIN POS terminal. This approach represents an early example of numeric voucher redemption but is limited to physical POS environments and cash-equivalent transactions. The present invention generalises the voucher concept into a digital-native authorization layer, in which the code establishes a verifiable link between a payer’s wallet and an on-chain transaction. It enables the same human-friendly entry method to initiate settlements, while remaining compatible with legacy terminals. The protocol thus bridges low-tech usability with high-integrity cryptographic settlement.

US 9489673 B2 presents a payment system in which a temporary numeric token is entered at a POS terminal to complete a purchase. This demonstrates user familiarity with short-lived codes but remains dependent on POS hardware and banking networks. The proposed invention generalises the voucher mechanism beyond purely terminal-based redemption by enabling code generation, transmission and settlement across peer-to-peer, e-commerce and wallet-based

contexts. Terminals may be used as one of several entry points into the voucher redemption flow, but unlike the system described in US 9489673 B2, the operation of the protocol is not dependent on their presence.

US 2017/0109718 A1 concerns secure cardless cash withdrawals in which a one-time code is generated and entered at an ATM. Although similar in its use of temporary numeric tokens, it is confined to physical cash withdrawals. The present invention extends the voucher concept to all classes of digital asset transfer, including deposits, withdrawals, merchant payments, direct wallet-to-wallet exchanges and interactions with smart contracts and decentralized programs, thus enabling wider scale of adoption.

US 9659294 B2 (Paydiant) describes a mobile phone ATM processing method in which a user associates a mobile device with a transaction via a token or code. This approach still depends on ATM infrastructure and QR or numeric entry. The invention described herein introduces out-of-band association through digital communication only, where the voucher serves as an authorization proxy rather than an identity token. The result is a fully software-defined, infrastructure-independent mechanism that directly resolves to blockchain-native settlement instructions, enabling seamless interoperability between wallets and decentralized networks while eliminating the dependency on proprietary hardware.

PL 401566 A1 discloses a method and system for authorising payments via a mobile device in a banking context, again reflecting OTP-based mobile authorization flows. The present invention departs from this model by embedding both authentication and transaction intent within a single voucher object that can be resolved directly into a blockchain-executable transaction. Each voucher thus generates a verifiable and auditable transaction record linking the authorization phase with the on-chain execution, creating a complete, tamper-evident trail of payment authorization and settlement—without reliance on banking infrastructure or issuer relationships.

In summary, the reviewed prior art demonstrates the longstanding use of short-lived numeric codes and vouchers for authorization in centralized payment environments. However, all such systems rely on stateful network connections, issuer or acquirer infrastructure, and banking-grade settlement frameworks. None of the existing solutions disclose or suggest a stateless voucher protocol that enables secure payer-to-payee association and transaction authorization within a

decentralized digital asset network. The invention described in this document addresses this deficiency by introducing a cryptographically verifiable, infrastructure-independent voucher mechanism that preserves the simplicity of numeric codes while enabling their direct use in digital asset transactions, thereby achieving secure, inclusive and intuitive value transfer for all types of users.

Summary Of The Invention

The present invention provides a method and system for conducting digital asset transactions through the use of short-lived digital vouchers that act as proxy authorizations between a payer and a payee. The invention addresses several technical and usability challenges identified in the background, including the complexity of key management, the need for direct and stateful wallet–application connections, and the lack of a simple, low-friction authorization mechanism.By introducing an ephemeral voucher session layer, the invention allows secure, low-friction transactions to be initiated and completed without exposing identities, public or private keys or requiring users to maintain persistent connections with external services.

In one embodiment, a payer initiates a transaction by generating a unique, short-lived voucher through a wallet client. The voucher is cryptographically bound to the payer’s authorization context but does not reveal the payer’s public or private key. It may be transmitted to the payee through any communication channel, including digital interfaces, in-person interaction, or through existing payment terminals. The payee redeems the voucher by submitting it to a payment server, which acts as an ephemeral session broker. The server validates the authenticity of the redemption request and resolves the voucher to an invoice describing the requested transaction. The payer receives a notification, reviews the transaction details, and authorizes the payment through their wallet. The resulting transaction is then executed and settled through a decentralized digital asset network.

The invention achieves strong cryptographic integrity through several technical mechanisms. First, communication between the wallet client and the payment server is secured end-to-end. In addition, messages originating from the payer include cryptographic signatures generated directly by the payer’s private key. This guarantees that all requests and authorisations originate from valid cryptographic identities, enabling the server to validate the authenticity and

integrity of every session. Second, each voucher is redeemed into a standardized blockchain-compatible transaction object. This ensures full backward compatibility with existing blockchain wallet interfaces and settlement mechanisms. Third, the invention abstracts the transaction creation process while remaining fully compatible with any blockchain interaction type, including smart contract calls. As a result, the voucher mechanism can be used not only for simple transfers but also for authorising and triggering complex on-chain operations without altering existing blockchain protocols.

Unlike prior voucher or one-time code payment systems, which are tightly coupled to centralized payment processors or issuer–acquirer infrastructures, the present invention defines a transport-agnostic protocol that supports both peer-to-peer and terminal-assisted redemption but is not dependent on either. The payment server acts only as a short-lived session coordinator and does not take custody of assets or execute financial transactions directly. Settlement occurs through existing decentralized ledger infrastructure, ensuring security and transparency consistent with blockchain trust models.

The combination of human-friendly voucher-based initiation, cryptographically secured communication, and seamless blockchain compatibility reduces complexity and lowers the barriers to entry for new users. At the same time, the system preserves compatibility with the existing ecosystem of blockchain wallets, transaction formats, and smart contract platforms. This enables a single authorization model to be used across peer-to-peer transfers, e-commerce transactions, deposit and withdrawal flows, and smart contract interactions, providing both improved accessibility for users and interoperability for developers and payment providers.

Through these technical features, the invention achieves a dual effect: it simplifies the user experience while maintaining or exceeding the cryptographic assurances offered by conventional wallet–dApp interactions, and it establishes a voucher-based authorization flow that is compatible with any blockchain settlement layer. This results in a secure, extensible and backward-compatible transaction authorization mechanism that overcomes the shortcomings of existing systems.

Brief Description Of The Drawings

The invention is described, by way of example only, with reference to the following drawings in which:

Figure 1 is a schematic diagram depicting the Voucher Generation and Transmission phases for voucher implementation according to this invention

Figure 2 is a schematic diagram depicting the Redemption, Authorization and Settlement phase for voucher implementation according to this invention

Detailed Description

Terminology

As used herein, the following terms have the meanings indicated below:

A ‘payment network’ is understood to refer to a network which is built around a centralized set of unified servers and/or distributed components (‘nodes’) that are distinct from one another. A payment network can function as a ledger and provide the capability of owning any type of asset, including a digitized form of currency or tokenized asset, and of transferring those assets freely to other users of that payment network.

A ‘cryptocurrency’ is understood to refer to a digitally native asset that functions as a medium of exchange, store of value, or unit of account within a payment network. A cryptocurrency typically operates on a cryptographically secured distributed ledger, often implemented via blockchain technology, and may be issued by a decentralized protocol, an organization, or another form of authority. Examples include Bitcoin, Ether, or stablecoins pegged to external assets. The term cryptocurrency may be used interchangeably with digital asset, tokenized currency, or digitized currency.

A ‘cryptocurrency network’ is understood to refer to a payment network that facilitates operations with cryptocurrencies or other digital assets, primarily enabling payments or transfers between users of the network — for example, between merchants and private individuals, or between individual users. However, the invention is not limited to simple payment transfers, as other operations such as financial products, decentralized finance (DeFi) services, or tokenized asset management can also be provided by a payment network operating in accordance with the invention. The invention particularly finds utility in account-based or key-

pair-based cryptocurrency networks, in which users each have an account or wallet that uniquely identifies them, typically by associating a public address with a corresponding private key. A cryptocurrency network can transfer digital assets between users but is not limited to this and may also enable other operations such as staking, lending, or automated contract execution.

A cryptocurrency network according to the invention can operate according to any currently known or future-developed techniques. The cryptocurrency network enables the handling of assets such as cryptocurrencies and related financial products. The phrase handling of assets is specifically understood to include any one or more of the execution, recording, verification, validation, and all post-enrollment activities related to the assets and their subset elements.

A ‘smart contract’ is understood to refer to a contract that is written in computer code. The smart contract code can be executed to produce a result; for example, a smart contract may enable one user to transfer funds to another user in a payment network. Smart contracts per se are known in the art and so a detailed description of smart contracts is not provided here in the interests of brevity. The term smart contract is understood to include an online smart contract that is deployed on the payment network’s ledger itself (e.g. a distributed ledger smart contract), as well as an external module with the functionality of a smart contract. The external module could be either software- or hardware-based.

A ‘payment’ is to be understood equally as a transaction and implies the creation of a new entry in the payment network’s or cryptocurrency network’s ledger indicating the transfer of assets between a payor and a payee. It is equivalent to a bank transfer but relates to a specific payment network and/or cryptocurrency network.

A ‘device’, or equivalently a ‘computer’, is understood to refer to an electronic piece of equipment that is capable of processing digital data. Examples include a mobile phone, a tablet, a laptop computer, a desktop computer, a gaming console, a smart TV, a cloud computing resource, and the like. Such devices typically include a processor, a memory, and a network interface capable of transmitting and receiving data from an external endpoint (e.g. another device or a component of a cryptocurrency network). Virtual machines are included within the scope of a device or computer. A user device is a device that is owned by, or otherwise under the control of, a particular user.

A ‘payer’ is understood to refer to a user or entity that initiates a transaction within a payment or cryptocurrency network. The payer is the party responsible for authorising the transfer of digital assets and therefore operates a wallet that holds the necessary cryptographic credentials, including a private key. In the context of this invention, the payer is the sole party capable of generating a digital voucher and subsequently authorising its redemption through transaction signing. The payer may be an individual, a business, or an automated agent acting on behalf of another entity.

A ‘payee’ is understood to refer to a user or entity that receives a payment within a payment or cryptocurrency network. Unlike the payer, the payee is not required to hold or operate a cryptographic wallet in order to receive funds within the system described by this invention. The payee may interact through a terminal, a merchant backend, or other compatible interfaces. The payee uses the voucher provided by the payer to initiate redemption, resulting in the generation of a transaction object for authorization and settlement.

A ‘wallet’ is understood to refer to a software or hardware system that allows a user to hold cryptographic credentials, sign messages or transactions, and interact with a payment or cryptocurrency network. Wallets typically store or control private keys corresponding to public addresses on the network. In the context of this invention, a wallet may also support additional functions, including the generation of digital vouchers and the signing of voucher-related messages exchanged with the payment server.

A ‘payer’s wallet’ is understood to refer to the specific wallet operated by the payer to generate digital vouchers, receive transaction invoices, and authorize blockchain transactions. The payer’s wallet is the only component in the system capable of producing cryptographic signatures that bind a voucher to an authorization context and enable subsequent retrieval and settlement of the associated transaction object.

A ‘payee interface’ is understood to refer to any hardware or software interface through which the payee receives a voucher and initiates redemption. This may include, but is not limited to, POS terminals, merchant systems, web interfaces, mobile applications, or other input devices capable of communicating the voucher to the payment server. The payee interface does not require access to private keys or cryptographic signing capabilities.

A ‘payment server’ is understood to refer to a network service that coordinates voucher lifecycle management without participating in the custody or execution of asset transfers. The payment server receives and validates voucher generation and retrieval requests, reserves and unreserves voucher codes, manages short-lived session records, and provides transaction objects to authorised payers for signing. The payment server has no access to private keys and does not initiate blockchain transactions on its own.

A ‘voucher’ is understood to refer to a short-lived, human-readable code or token generated by the payer’s wallet and associated with a specific authorization context. The voucher may consist of numeric or alphanumeric characters and serves as an ephemeral proxy for payment authorization. It can be transmitted between the payer and the payee through any supported channel without exposing cryptographic material. The voucher enables indirect, stateless interaction between payer and payee interfaces.

A ‘voucher code’ is understood to refer to the literal sequence of numeric or alphanumeric characters that uniquely identifies a voucher within its validity period. A voucher code may be entered, spoken, scanned, transmitted, or otherwise conveyed to the payee, and subsequently submitted to the payment server for redemption. Voucher codes are reserved at the time of generation and unreserved upon session completion.

A ‘voucher session’ is understood to refer to a short-lived data structure maintained by the payment server that associates a voucher code with the payer’s identity and transaction parameters. The voucher session exists from the moment a voucher is generated until the transaction object is retrieved by the authorised payer. Once retrieval occurs, the session is finalized and destroyed, and the voucher code is released back to the available pool.

A ‘session record’ is understood to refer to the stored representation of a voucher session on the payment server, including metadata such as voucher code, session identifiers, expiration time, payer binding, and any additional contextual parameters necessary to support invoice generation and retrieval. The session record does not contain private keys or signed transactions.

A ‘transaction object’ is understood to refer to a structured data representation of a transaction generated after voucher redemption. The transaction object contains all necessary information required for a blockchain transaction, such as recipient

address, asset type, amount, network identifier, and additional payload data (e.g., smart contract invocation parameters). The transaction object is transmitted to the payer's wallet for signing and subsequent submission to the blockchain.

An 'invoice' is understood to refer to the intermediate representation of a transaction request generated when a payee redeems a voucher. The invoice is produced by the payment server based on payee-provided details and the existing voucher session. It is not a signed transaction but a pending authorization object, which the payer may fetch, review, and authorize through the wallet. Upon authorization, the invoice is transformed into a signed transaction object for settlement on the blockchain network.

Solution

The present invention provides a method and system enabling secure authorization and conducting digital asset transactions using short-lived vouchers. These vouchers serve as ephemeral authorization proxies between a payer and a payee. The system allows transactions to be initiated and authorised without exposing public or private keys, maintaining persistent wallet-to-application connectivity, or relying on issuer–acquirer payment infrastructures. It is designed to be compatible both with fully digital environments and existing payment terminals, while ensuring cryptographic integrity and interoperability with existing blockchain protocols.

In the system according to the invention, the payer operates a wallet client capable of generating, signing and transmitting voucher-related requests. The wallet can be a software application, a hardware device or an integrated component of a payment platform. The payee may also operate a wallet client capable of receiving and redeeming vouchers. However, unlike conventional digital asset payment systems, operation of a wallet by the payee is not required. This design allows the payee to interact with the system through low-tech interfaces such as standard payment terminals or merchant backend systems, without needing to manage cryptographic keys or blockchain addresses. The cryptographic authorization of the transaction remains anchored in the payer's wallet, while the payee simply initiates a voucher redemption through an input interface.

A payment server provides a coordination layer that manages the temporary lifecycle of vouchers and links payer and payee sessions. The server does not act as a financial intermediary and does not control or transfer digital assets directly; instead, it stores minimal, short-lived state data necessary to associate a voucher with a payer until the voucher expires or is redeemed.

A voucher is a short-lived, human-friendly code generated by the payer's wallet at the time of initiating a transaction. Each voucher is uniquely associated with a specific authorization context at the time of its creation. It can be transmitted between parties using any available channel, including digital interfaces, voice, NFC, QR code, or entry through a payment terminal. The voucher has a defined time-to-live (TTL) parameter that limits its validity period. In order to support a wide range of device capabilities and terminal types, the system operates in two modes: a numeric mode and an alphanumeric mode.

In numeric mode, the voucher consists of a short sequence of digits, for example six digits, making it compatible with a wide range of existing payment terminals, including legacy POS devices and standard ISO/IEC 7812-1 and ANSI X4.13 compliant keypads that accept only numeric input. Examples of compatible devices include stand-alone countertop payment terminals, integrated POS systems, unattended payment kiosks, ATMs, and older keypad-based merchant terminals. These devices allow the payer to communicate the numeric voucher quickly, either verbally or by manual entry, ensuring broad compatibility even in low-tech retail environments.

In alphanumeric mode, the voucher consists of a short sequence of letters and digits (for example six alphanumeric characters). This mode significantly increases the number of unique vouchers that can be generated in parallel, thereby improving scalability and reducing code-collision risk. Alphanumeric vouchers may be entered on or scanned through interfaces that support alphanumeric input, for instance: modern POS terminals with full QWERTY or hybrid keypads (which comply with keyboard layout standards such as ISO/IEC 9995-2:2009 'Alphanumeric section') and touchscreen POS terminals; smart-POS devices and merchant web or mobile dashboards which allow alphanumeric keyboard entry; and customer self-checkout kiosks with software-based alphanumeric entry. Example interfaces include smart POS terminals (e.g., touchscreen POS models with on-screen alphanumeric entry), mobile payment

apps, merchant dashboard portals and web checkout fields that accept alphanumeric voucher codes.

The selection of voucher mode can be determined dynamically based on terminal capabilities, merchant configuration or user preference. For example, when interacting with a numeric-only POS terminal, the payer’s wallet will generate a numeric voucher to ensure compatibility. When interacting with an alphanumeric-capable device or a digital interface, the payer’s wallet can generate an alphanumeric voucher, providing increased scalability and enhanced security. By supporting both modes, the invention maintains interoperability with low-tech infrastructure while enabling high-volume operation in modern environments.

All communications between the wallet clients and the payment server are secured end-to-end. In addition to the secure communication channel, messages transmitted from the payer’s wallet to the payment server include cryptographic signatures generated using the payer’s private key. These signatures are applied at the application level and provide an additional layer of security by establishing a verifiable link between the payer and the voucher within a specific session. This ensures that the payment server can unambiguously associate the voucher with the correct payer context, prevents unauthorized parties from querying or interfering with other active sessions, and guarantees the integrity of the payer’s authorization intent. This cryptographic binding is not required on the payee side, since the transfer relies entirely on the payer’s authorization.

Algorithm Description

When a payment is initiated, the payer’s wallet begins by generating a voucher and establishing a corresponding session on the payment server. This generation step is the starting point of the transaction flow and defines the association between the payer’s cryptographic identity, the voucher code, and the transaction parameters. The wallet submits a voucher generation request, which is validated by the payment server and stored as a short-lived session record. The server allocates the voucher, reserves it for the specified time-to-live, and returns the code and session metadata to the payer. This process creates the binding between a human-readable voucher and the underlying authorization context, enabling the payee to later redeem the code without requiring access to the payer’s wallet or cryptographic material. From this point onward, all subsequent steps of the transaction flow operate with reference to this generated voucher.

After the voucher is generated, it is transmitted from the payer to the payee through a transport-agnostic mechanism. The system does not prescribe a single communication channel; instead, the voucher may be delivered by any medium appropriate to the context of the transaction. This may include direct digital communication, such as copy-and-paste in a mobile or web application, instant messaging, or embedded deep links. It may also include physical or local transfer mechanisms such as the payer reading the voucher aloud for manual entry, displaying the code on their screen for the payee to type or scan, or encoding the voucher in a QR code or NFC payload. When the transaction is initiated through a payment terminal, the payer may enter the voucher directly on the terminal keypad or present it for scanning where supported.

Once the payee receives the voucher through any of the supported transmission methods, it may initiate the redemption process. The payee provides the voucher to its chosen interface, which may include a payment terminal, merchant backend system, web interface, or mobile application. The interface then communicates the voucher to the payment server, triggering the verification and invoice generation procedures, in which the voucher transitions from a shared code to a structured, verifiable payment request.

The voucher is then transformed into a standardized transaction object that is directly compatible with existing blockchain protocols. This transformation process constitutes a key technical differentiator between the present invention and traditional numeric voucher systems. Rather than mapping the voucher to an account or clearing request in a centralized payment network, the system resolves the voucher into a structured transaction that can be executed on a decentralized ledger.

The transaction object is created based on the information provided by both the payer and the payee during the voucher session, including payment amount, asset type, target account or smart contract address, network identifier, and any required execution parameters. The resulting object conforms to the expected format of the target blockchain, ensuring that the payer’s wallet can sign and broadcast it without requiring protocol extensions or proprietary infrastructure.

For example:

- On the Ethereum network, the voucher may be resolved to a standard transaction object compatible with the JSON-RPC interface as defined by EIP-1193. This allows the wallet to invoke methods such as eth_sendTransaction or eth_signTransaction using its existing signing logic. The transaction may include simple value transfers (e.g., ETH or ERC-20 transfers) or arbitrary smart contract interactions via to, data, and value fields. This ensures full compatibility with the existing Ethereum ecosystem, including browser-based wallets, hardware wallets, mobile wallets, and dApps.
- On the Bitcoin network, the voucher may be resolved into a transaction following the Bitcoin raw transaction format. The payment server provides the payer's wallet with the necessary transaction outputs, including recipient address and amount, while leaving inputs and signature generation to the wallet. The wallet then signs the transaction with the payer's private key, producing a valid Bitcoin transaction that can be broadcast through any compatible node or service.
- On the Solana network, the voucher is resolved to a transaction structure consistent with the Solana Web3 interface. The payment server supplies transaction metadata such as the target public key, lamport amount, and instruction set. The payer's wallet signs the transaction using the standard Solana signing procedure, enabling settlement through any compatible node endpoint.

In addition to simple transfers, the redemption mechanism can be used to authorize any interaction compatible with the underlying blockchain. This includes but is not limited to smart contract method invocations, token approvals, staking operations, or cross-chain bridge interactions. By separating authorization and transaction construction from transport and UI constraints, the system enables a unified, human-friendly authorization layer while remaining fully interoperable with existing blockchain standards.

Following voucher redemption and generation of the corresponding transaction object, the payment server issues a notification to the payer's wallet indicating that a pending invoice is available for authorization. The payer's wallet initiates a secure retrieval process by generating a signed request to the payment server using the cryptographic identity previously associated with the voucher session.

This ensures that access to the transaction object is restricted exclusively to the authorised payer.

Upon receiving the signed retrieval request, the payment server validates the signature against the payer identity bound to the session. If verification is successful, the server returns the transaction object to the payer's wallet. Immediately thereafter, the server finalises the voucher lifecycle by marking the session as completed, unreserving the associated voucher code, and returning it to the available code pool for reuse in subsequent sessions. The session record is then securely destroyed or marked inactive to prevent replay or double-spend scenarios. This guarantees that each voucher may only be redeemed and retrieved once within its validity period.

The transaction object is then reviewed and authorised by the payer through the wallet's native signing mechanism in accordance with the protocol of the target blockchain network. For example, in an Ethereum-compatible environment, the wallet may utilize EIP-1193 methods; in a Bitcoin environment, it may apply standard raw transaction signing; and in a Solana environment, it may employ the network's native signing format. After authorization, the wallet transmits the signed transaction to the blockchain network, initiating settlement through the chain's native consensus process.

Because the invention abstracts only the authorization and initiation phase, it does not modify or interfere with the native settlement process of any blockchain. Each transaction is ultimately signed by the payer's wallet using its existing cryptographic key pair, ensuring full compatibility with current protocols. This architecture allows the same voucher mechanism to support multiple transaction types, including peer-to-peer transfers, token operations, deposits and withdrawals, merchant payments, and smart contract function invocations.

This design also ensures that no trusted intermediary is required to execute the transaction. The payment server stores a minimal set of session attributes for each voucher, including the voucher identifier, payer session reference, and expiration metadata. It does not store or process private keys and has no ability to initiate blockchain transactions on its own or forge transactions. This design ensures a loose coupling between the voucher coordination layer and the blockchain settlement layer.

In this way, the invention addresses multiple shortcomings of prior systems identified in the background. It simplifies the user experience by abstracting away complex wallet-to-dApp authorization steps, eliminates the requirement for persistent sessions, and provides a simple voucher-based interface for initiating secure transactions without exposing public or private keys. At the same time, it preserves cryptographic integrity through signed communications and direct wallet-based settlement. It is compatible with both existing terminal infrastructure and other digital channels, ensuring broad applicability and ease of integration.

Algorithmic Flow of the Invention

1. **Voucher generation phase**
    - **S101:** Payer initiates a digital payment in a wallet client (processor, secure key storage, network interface); Wallet builds VOUCHER_INIT (JSON/CBOR): payer identifier (hashed address/session token), optional context (amount/currency/merchant), timestamp, nonce.
    - **S102:** Wallet signs VOUCHER_INIT with payer private key (e.g., ECDSA/Ed25519).
    - **S103:** Sends over secure transport (e.g., HTTPS/TLS 1.3 or gRPC).
    - **S104:** Payment server verifies signature, timestamp, nonce.
    - **S105:** Server generates unique voucher code (e.g., 6-char alphanumeric via CSPRNG).
    - **S106:** Server creates a session record (voucher, payer ref, TTL, context hash) in storage (e.g., SQL).
    - **S107:** Server returns VOUCHER_ISSUE (voucher, TTL, session metadata).
    - **S108:** Wallet displays the voucher.
2. **Voucher transmission phase**
    - **S109:** Payer transmits the voucher by any channel (manual numeric entry — ISO/IEC 7812-1 / ANSI X4.13; alphanumeric entry — ISO/IEC 9995-2; QR; NFC; BLE; messaging).
3. **Voucher redemption and invoice generation**

- **S201:** Payee interface captures voucher and builds VOUCHER_REDEEM with voucher code + payee txn parameters (amount, asset, destination, network).
- **S202:** Payee sends VOUCHER_REDEEM over secure transport.
- **S203:** Server looks up active session by voucher; verifies TTL not expired.
- **S204:** Server constructs a blockchain-compatible *unsigned transaction object* (TX_UNSIGNED) for the target network using the session and payee parameters (e.g., destination, asset/type, amount, nonce placeholder, fee/gas params template, optional contract call data), without applying any signature or private-key operation.
- **S205:** Server marks session state as *pending-signature* and sends TX_PENDING_NOTIFY to the payer's wallet (WebSocket/push/MQTT/AMQP) indicating TX_UNSIGNED is ready for retrieval.

4. **Invoice retrieval and payer authorization**

- **S206:** Wallet receives notification and prepares TX_FETCH with voucher/session ref, signed by the payer's private key.
- **S207:** Server verifies the signature against the payer identity bound to the session and returns the TX_UNSIGNED object.
- **S208:** Server finalises the session: marks *completed*, deletes session metadata, unreserves the voucher code and returns it to the available pool.
- **S209:** Wallet displays transaction details (amount, asset, destination, fees/network).
- **S210:** If permitted by the protocol, the wallet may fill in/adjust dynamic fields (e.g., nonce, max fee, priority) before/while signing, consistent with network rules.
- **S211:** Upon payer confirmation, the wallet validates TX_UNSIGNED and authorizes it by signing (transforming the unsigned transaction into a signed transaction object, TX_SIGNED).

5. **Transaction signing and settlement**

- **S212:** Wallet broadcasts TX_SIGNED via the network's native interface (e.g., JSON-RPC/Web3 provider, REST, or node API).
- **S213:** Blockchain processes and settles the transaction per its consensus;
- **S214**: wallets update balances/status from on-chain confirmations.

The above sequence is depicted in [FIG 1] and [FIG 2].

In the exemplary embodiment described, a series of message types are defined to facilitate communication between the payer wallet, the payment server, and other system components. These messages include, for example, VOUCHER_INIT (voucher generation request), VOUCHER_ISSUE (server response containing voucher and session data), VOUCHER_REDEEM (voucher redemption request from payee), TX_PENDING_NOTIFY (notification of pending invoice to payer), TX_FETCH (payer request to retrieve invoice and transaction data), TX_UNSIGNED and TX_SIGNED (unsigned and signed transaction objects, respectively). Each message represents a structured communication between system components that conveys specific transactional state and context information. The detailed syntax, encoding, or transport of these messages is not critical to the invention and may vary across implementations. For example, messages may be expressed as JSON, CBOR, Protocol Buffers, or another serialization format, and may be exchanged via HTTPS, gRPC, WebSocket, or other secure transport mechanisms. The essential aspect is the logical sequencing and purpose of the messages in establishing, authorising, and completing the voucher-based transaction flow as disclosed herein.

The above sequence illustrates one embodiment of the invention; variations are possible. For example, the different blockchain platforms or standards than one provided in example sections can be adapted or the communication between wallets and the payment server can be performed through different transport layers, including encrypted network protocols or terminal interfaces etc.

Claims

1. A computer-implemented method of authorising and settling a digital asset transaction within a payment or cryptocurrency system, the method comprising:

generating, by a payer wallet, a short-lived voucher code associated with a transaction authorization context;

transmitting the voucher code from the payer to a payee via a transport-agnostic communication channel;

redeeming, by a payee interface, the voucher code through a payment server, thereby triggering generation of a transaction invoice;

notifying the payer wallet of the pending transaction invoice;

retrieving, by the payer wallet, the transaction invoice from the payment server using a cryptographically signed request;

finalising, by the payment server, the voucher session and unreserving the voucher code;

authorising, by the payer wallet, the retrieved transaction invoice and generating a signed transaction object compatible with a blockchain settlement layer; and

broadcasting the signed transaction object to the blockchain settlement layer for processing and settlement.

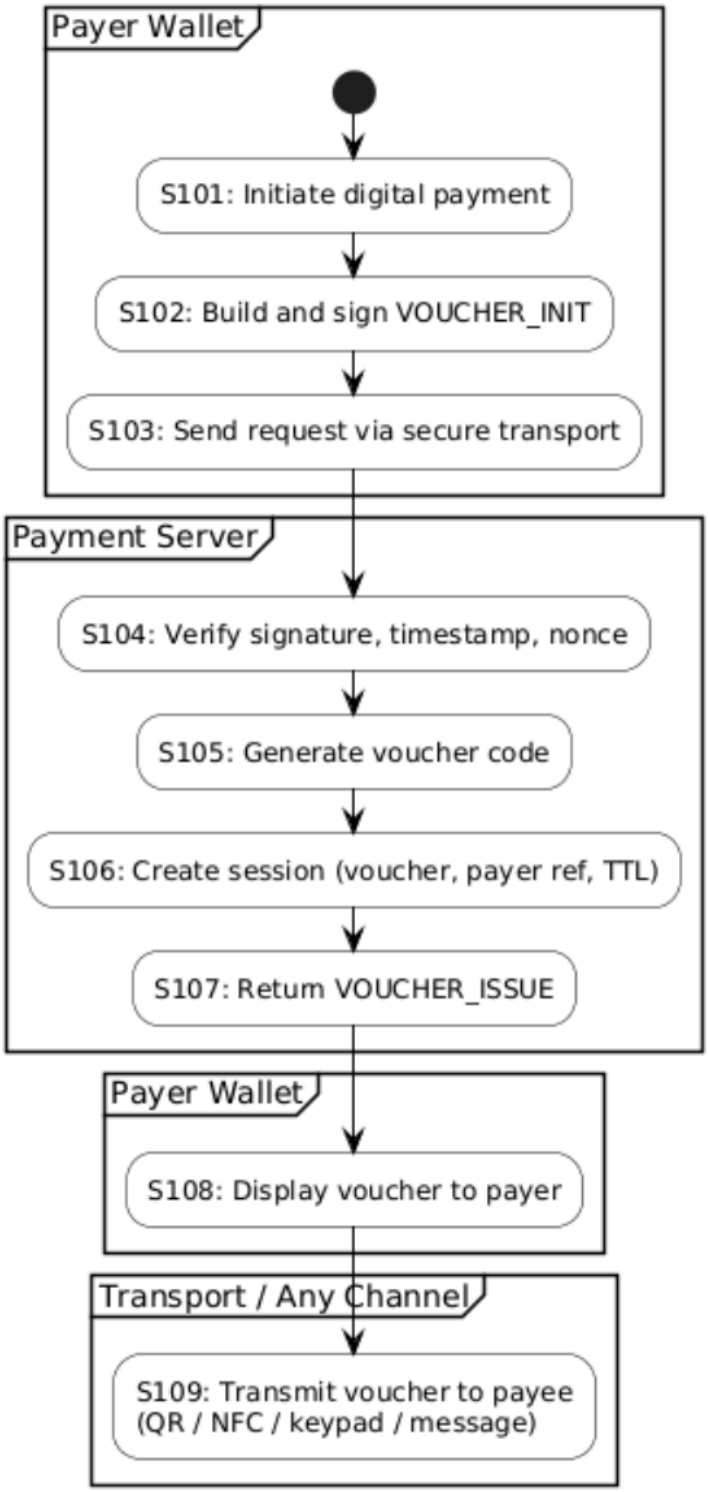
2. The method of claim 1, wherein the voucher is generated in either numeric or alphanumeric format, the format being determined dynamically based on terminal capability or user configuration.
3. The method of claim 1, wherein the payment server allocates the voucher code, creates a short-lived session record, reserves the code for a defined time-to-live, and releases the code back to a shared pool upon session finalisation.
4. The method of claim 1, wherein the voucher redemption process is executable through a payee interface lacking cryptographic signing capabilities.
5. The method of claim 1, wherein the payment server does not store or process private keys and is not capable of initiating or forging blockchain transactions.

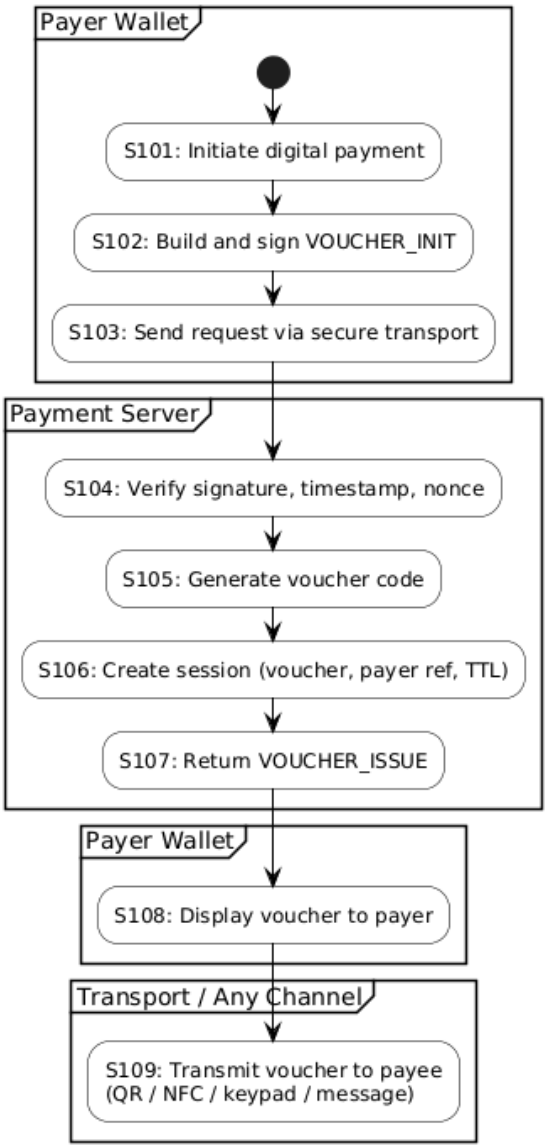
Abstract

The invention relates to a method and system for authorising digital asset transactions using short-lived digital vouchers acting as proxy authorizations

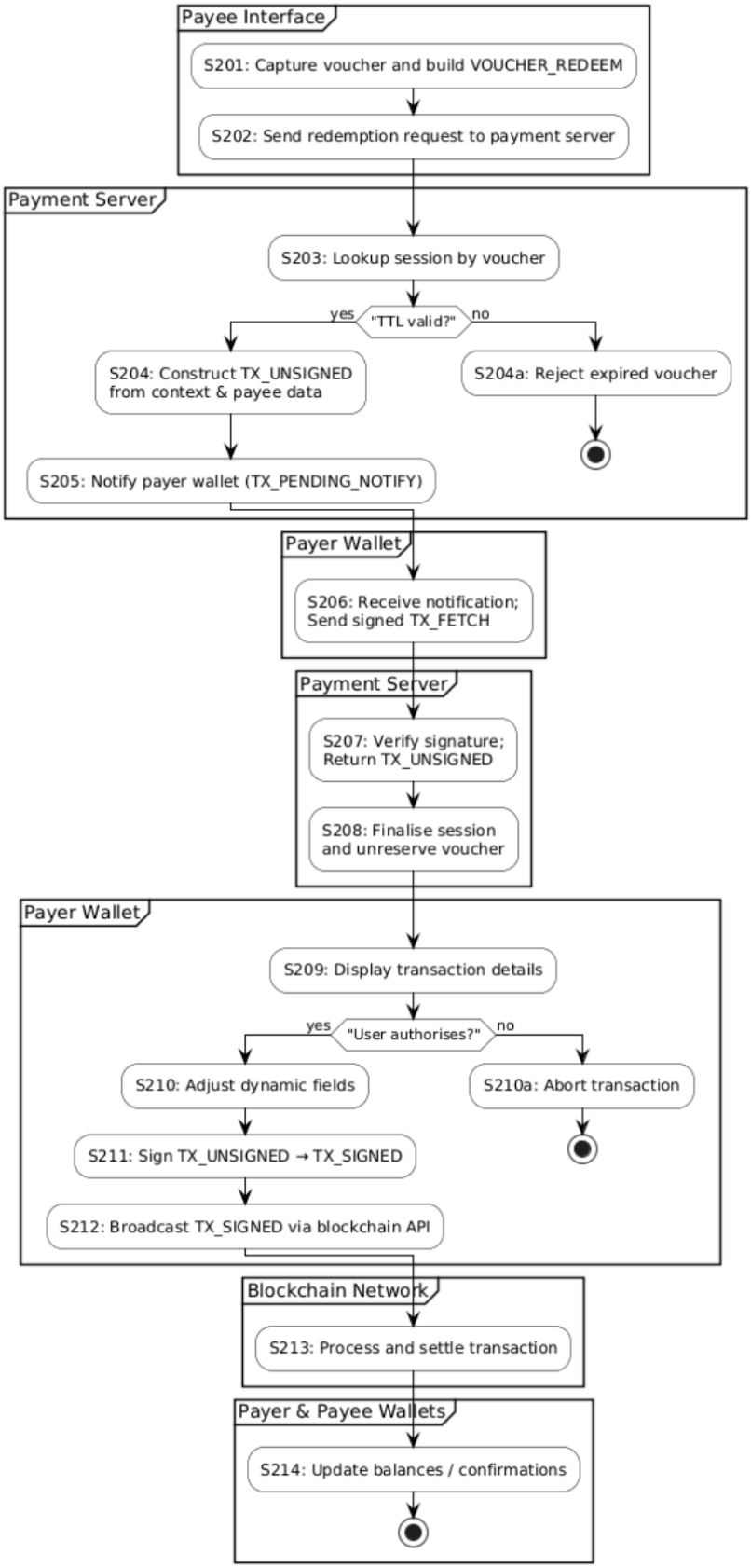
between a payer and a payee. A wallet client generates a unique voucher cryptographically linked to the payer’s authorization context without revealing the private key or wallet address. The voucher may be transmitted through any communication channel, including digital networks or existing payment terminals, and is redeemed by the payee via a payment server functioning as an ephemeral session broker. Communication with the server is secured through signatures generated by wallet clients, ensuring cryptographic authenticity and integrity. Upon redemption, the voucher is resolved into a standardized blockchain transaction compatible with existing protocols, enabling backward-compatible settlement on decentralized networks. The system abstracts transaction generation while supporting any blockchain interaction, including smart contract calls. This approach simplifies user experience, reduces reliance on persistent wallet connections and issuer–acquirer infrastructure, and enables secure, low-friction digital asset payments across peer-to-peer, e-commerce and terminal-assisted contexts.

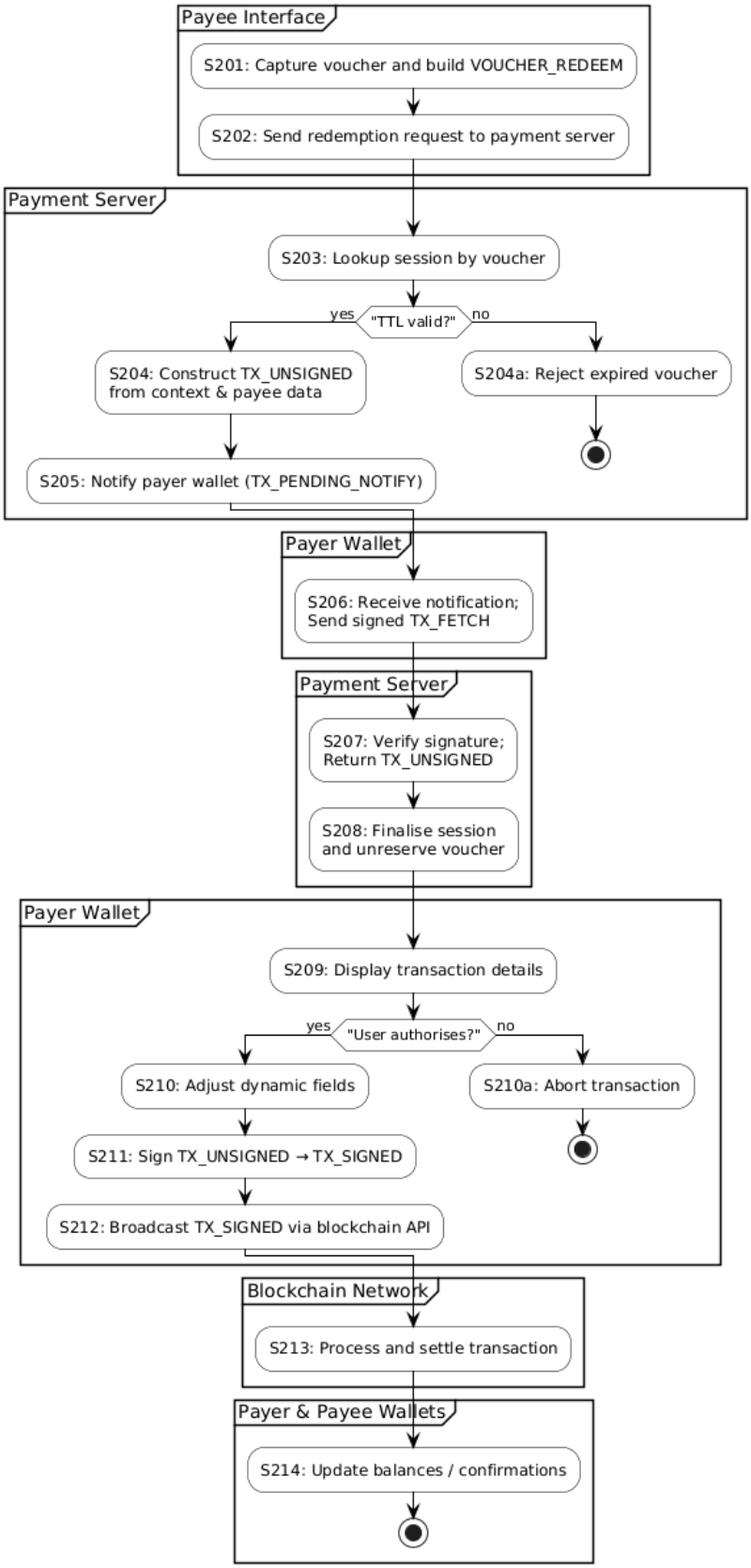
Drawings


Payer Wallet
S101: Initiate digital payment
S102: Build and sign VOUCHER_INIT
S103: Send request via secure transport
Payment Server
S104: Verify signature, timestamp, nonce
S105: Generate voucher code
S106: Create session (voucher, payer ref, TTL)
S107: Return VOUCHER_ISSUE
Payer Wallet
S108: Display voucher to payer
Transport / Any Channel
S109: Transmit voucher to payee
(QR / NFC / keypad / message)



[FIG 1] Voucher Generation and Transmission Phase (S101–S111)


Payee Interface
S201: Capture voucher and build VOUCHER_REDEEM
S202: Send redemption request to payment server
Payment Server
S203: Lookup session by voucher
"TTL valid?"
yes
no
S204: Construct TX_UNSIGNED from context & payee data
S204a: Reject expired voucher
S205: Notify payer wallet (TX_PENDING_NOTIFY)
Payer Wallet
S206: Receive notification; Send signed TX_FETCH
Payment Server
S207: Verify signature; Return TX_UNSIGNED
S208: Finalise session and unreserve voucher
Payer Wallet
S209: Display transaction details
"User authorises?"
yes
no
S210: Adjust dynamic fields
S210a: Abort transaction
S211: Sign TX_UNSIGNED → TX_SIGNED
S212: Broadcast TX_SIGNED via blockchain API
Blockchain Network
S213: Process and settle transaction
Payer & Payee Wallets
S214: Update balances / confirmations


Payee Interface
S201: Capture voucher and build VOUCHER_REDEEM
S202: Send redemption request to payment server
Payment Server
S203: Lookup session by voucher
"TTL valid?"
yes
no
S204: Construct TX_UNSIGNED from context & payee data
S204a: Reject expired voucher
S205: Notify payer wallet (TX_PENDING_NOTIFY)
Payer Wallet
S206: Receive notification; Send signed TX_FETCH
Payment Server
S207: Verify signature; Return TX_UNSIGNED
S208: Finalise session and unreserve voucher
Payer Wallet
S209: Display transaction details
"User authorises?"
yes
no
S210: Adjust dynamic fields
S210a: Abort transaction
S211: Sign TX_UNSIGNED → TX_SIGNED
S212: Broadcast TX_SIGNED via blockchain API
Blockchain Network
S213: Process and settle transaction
Payer & Payee Wallets
S214: Update balances / confirmations

[FIG 2] Redemption, Authorization and Settlement Phase (S201–S214)

Original file → https://docs.google.com/document/d/1LN-NmkqLY6pvYec-D3wWPkhwXAq1lqWmUxuipE2c2FE